



**A**  
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**May, 2026**

## **DECLARATION**

We hereby declare that this submission is our own work and that, to the best of our knowledge and belief, it contains no material previously published or written by another person nor material which to a substantial extent has been accepted for the award of any other degree or diploma of the university or other institute of higher learning, except where due acknowledgment has been made in the text.

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## CERTIFICATE

This is to certify that Project Report entitled “Sustainable Energy Management in 5G Networks” which is submitted by Aayushker Singh, Akshat Srivastav, Anshika Agarwal, Pari Gupta in partial fulfillment of the requirement for the award of degree B. Tech. in Department of CSE (AI) of Dr. A.P.J. Abdul Kalam Technical University, Lucknow is a record of the candidates own work carried out by them under my supervision. The matter embodied in this report is original and has not been submitted for the award of any other degree.

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## **ABSTRACT**

The shifting to 5G networks, which is more energy-efficient per bit than 4G, raises serious concerns regarding the total power consumption, as it is being densely deployed and traffic is expected to increase. This energy consumption is dominated by the Radio Access Network (RAN), especially base stations, and is thus a prime area to focus on in optimization.

To overcome this problem, a complex energy-saving approach needs to be considered, which includes base station hardware, software, and operational environment upgrades, like transmission power management and sleep states. The energy footprint of a 5G cell site is determined by various interrelated factors: hardware efficiency, operating configurations, dynamic traffic patterns, and optimization methods such as scheduling and resource allocation.

Because of this complexity and variability, data-driven approach is necessary and proactive. It is essential to develop proper and stable models of energy consumption projection in order to have a sustainable and economically viable 5G implementation.

Machine learning (ML) can have a prominent role since it is able to analyze large amounts of operational data, including traffic patterns, environmental conditions, and configuration metrics, to support data-driven decisions. It also facilitates real-time network optimization by predictive adjustments and reinforcement learning to enable base stations to align capacity with demand and enhance energy efficiency without impacting quality of service (QoS).

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## LIST OF ABBREVIATIONS

|       |                                       |
|-------|---------------------------------------|
| RAN   | Radio Access Network                  |
| LTE   | Long Term Evolution                   |
| ITU   | International Telecommunication Union |
| MIMO  | Multiple Input Multiple Output        |
| LSTM  | Long Short Term Memory                |
| AI    | Artificial Intelligence               |
| DL    | Deep Learning                         |
| ML    | Machine Learning                      |
| CPU   | Central Processing Unit               |
| GPU   | Graphics Processing Unit              |
| BS    | Base Station                          |
| ID    | Identification                        |
| NaN   | Not a Number                          |
| MLP   | Multi Layer Perceptron                |
| CSV   | Comma-Separated Values                |
| MAE   | Mean Absolute Error                   |
| RMSE  | Root Mean Squared Error               |
| MAPE  | Mean Absolute Percentage Error        |
| $R^2$ | Coefficient of Determination          |

# CHAPTER 1

## INTRODUCTION

### 1.1 Introduction

The fast development of the wireless communication technologies has changed how the contemporary world communicates, socializes, and does business. The fifth generation (5G) of mobile networks is one of such impressive technological improvements, which promises to provide extremely high data rates, exceptionally low latency, a significant increase in reliability, and the ability to provide connections to large numbers of devices. The 5G, in contrast to other previous generations, is not confined to better mobile broadband; 5G is the heart of new technologies like the Internet of things (IoT), smart cities, autonomous vehicles, augmented and virtual reality, industrial automation, and smart healthcare systems. The 5G networks are emerging as an imperative part of the global infrastructure as countries and industries are increasingly becoming reliant on digital connectivity.

The implementation of 5G networks has also come with significant energy usage challenges despite all the benefits that the network has. Despite the fact that the 5G technology should be significantly more energy-saving per transmitted bit than the 4G as it can be estimated that 4G networks use almost four times less energy, the high-energy consumption of 5G networks is much higher. The main reasons that cause this increment include intensive coverage of base stations, the use of elevated frequency bands, work with wider bandwidths, and intricate signal processing to provide huge connectivity and sophisticated services. This is causing the overall power consumption of telecom networks to increase as opposed to going down.

Radio Access Network (RAN), particularly base stations contribute to a considerable chunk of this energy consumption. Research has shown that the total energy consumption of the

network is over 70 percent due to operations of the base stations. Though other parts of the whole consumption like data centers and transport networks have a role, their contribution to the overall consumption is relatively lesser. This demand is also augmented by the exponential growth of linked devices triggered by network IoT and intelligent infrastructure. It is now an urgent research and industrial issue due to the projections that telecommunications will contribute to a noticeable segment of global electricity usage in the next decade, which implies energy efficiency in 5G networks.

There are various impacts of high consumption of energy. Economically, the cost of energy is a significant part of operational expense (OPEX) to telecom operators, and can be as large as 20-40 percent of total expense. The profitability and affordability of services can be the direct result of unoptimized use of energy. Environmental wise, more consumption of electricity leads to greater carbon emissions, which are problematic to the global sustainability objectives and climate commitments. With governments, and international bodies highlighting the need towards sustainable development and climate action, and industries take to reducing their footprint on the environment, the telecommunications industry included, must go about it intelligently.

The 5G network sustainable energy management concept has acquired a considerable importance in this regard. Old methods of energy saved like the static configuration optimisation or hardware upgrades are not enough to deal with the dynamic and large scale demands of network environment. Rather, data-driven smart solutions are needed to comprehend network behavior and estimate the patterns of energy consumption as well as prescribe proactive optimization strategies.

Machine Learning (ML) has turned out to be a potent complex systems modeler and hidden insights in big data. ML methods in telecommunications have the ability to examine various parameters including network load, hardware configuration, transmission power, bandwidth allocation and time usage behaviour to predict energy consumption at high accuracy. Through predictive analytics, telecom operators will be able to shift towards proactive and optimally effective decision-making, as opposed to reactive energy management.

Hence, the incorporation of machine learning approaches to the 5G energy management framework is a new avenue that promises the creation of network solutions not only with high performance but also with low energy and environmental sustainability. The present project is aimed at overcoming this difficulty by keeping predictive models that can estimate and optimize energy use in 5G base stations.

## **1.2 PROJECT DESCRIPTION**

### **1.2.1 Problem Statement**

Massive implementation of the 5G networks has remarkably escalated the energy requirement of the telecom infrastructure, especially the base station. The 5G technology is more effective in terms of data transmission per unit of energy, but the necessity to install base stations at a dense level, use frequencies, and need the processing capabilities leads to a significant total power consumption. Over two-thirds of network energy expenditure falls in Radio Access Network and therefore base stations are the main contributors of the operational energy costs and carbon emissions.

The methods of current energy management tend to be based at the static rules or conventional optimization methods, which can not be adequate in processing the dynamic traffic variations of the networks and the dynamic interactions among the various configuration parameters. Optimization strategies can be useless or even counterintuitive without proper forecasting systems. Thus, a robust predictor system that is capable of estimating the energy usage under changing operating conditions and giving information toward smart decisions of saving energy is desperately required.

The issue that is tackled in this project is the issue of prediction of hourly energy consumption of 5G base stations using machine learning and through these predictions, they can support sustainable energy expansion plans.

### **1.2.2 Motivation**

This research is driven by the technological issues as well as environmental issues. Because 5G networks are increasingly becoming common all around the world, their aggregate energy consumption is very problematic in terms of sustainability. Without optimization, massive 5G implementation would dramatically raise the cost of operation by the telecommunication companies and will act as a catalyst to the escalation of carbon emission.

The concept of the project was based off of a real world data science challenge on an open research competition platform that was centered around creating AI-driven solutions in energy efficiency of 5G base stations. Real data sets on operational tasks opened up an ability to examine advanced machine learning methods to a practical and socially swaying issue.

Besides, this project is in line with a number of Sustainable Development Goals (SDGs) such as:

- Ensuring infrastructure energy efficiency (SDG 7 – Affordable and Clean Energy)
- Promoting industry and environmentally-friendly infrastructure (SDG 9)
- Smart and sustainable urban development (SDG 11) must be supported.
- Engaging in climate action through lessening carbon emissions (SDG 13)

All these reasons underscore the need to have smart and sustainable energy management provision in next-generation network.

### **1.2.3 Objectives**

The major objectives of this project are:

1. To create a machine learning structures-based prediction system that can clearly predict the hourly energy consumption of 5G base stations.
2. In order to monitor and determining factors that affect energy consumption such as network load, hardware setups, frequency, bandwidth, and transmitting power, and temporal behavior.



3. To compare and contrast various machine learning models, deep learning and ensemble models in particular, to establish which predictive method is the most effective.
4. To deploy a powerful Random Forest regression model, which has a good balance of accuracy, interpretability, and generalization performance.
5. To offer evidence-based energy-saving guidelines to be used in proactive optimization strategies in the 5G networks.

#### 1.2.4 Overview of Methodology and Model Selection

The project has taken an open research dataset that is availed as a part of an international data science competition. The data contains a sophisticated base station data regarding load levels, record of consumed energy, radio unit models, operating modes, frequencies, bandwidth, and amount of transmitted power. The time and base station identifiers were combined to create one analytical structure out of multiple datasets.

In offering a conceptual insight into the functionality of various operational and structural parameters, which affect power consumption in 5G base stations, Figure 1.1 elucidates the connection existing between system critical parameters and energy consumption.

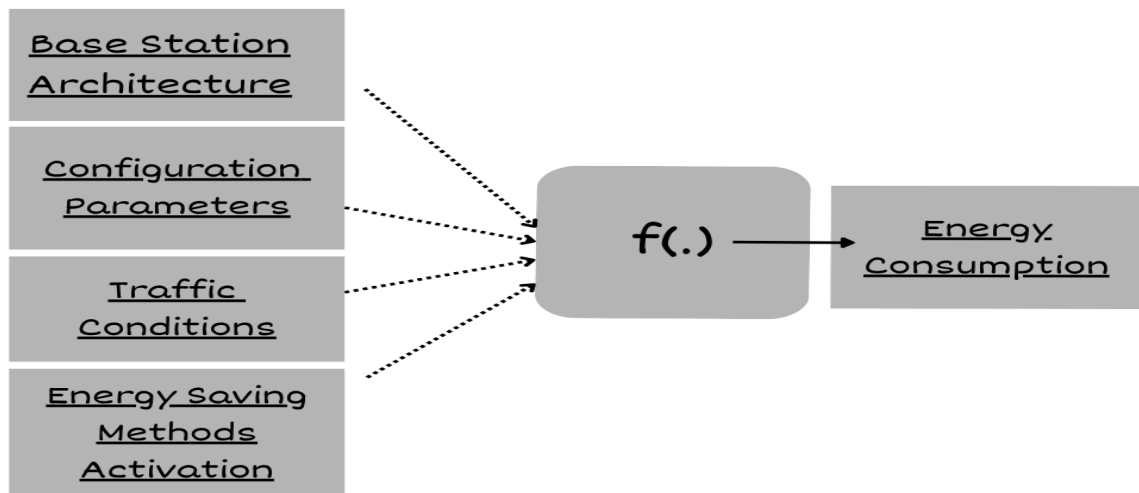


FIGURE 1.1: Conceptual Representation of Factors Influencing Energy Consumption in 5G Base Stations

The Figure 1.1 shows that the 5G base station energy consumption can be modeled as a functional of various interdependent components. These are base station architecture, configuration settings, traffic settings and enabling energy saving mechanisms. Hardware characteristics of the hardware used to design an architectural part include the antenna system and radio units. Such parameters as transmission power, operating frequency, and bandwidth allocation are included as configuration parameters. Traffic conditions are changes in the load of a network dynamically in time, whereas energy-saving techniques are operational modes that are applied during the reduction of unwarranted power consumption.

The function  $f(.)$  distinguishes the predictive modeling framework as a result of the interactions between these cause and effect that is quite complex and non-linear in nature and this is represented via  $f(.)$  in the figure. The model uses this information to learn the relationship between different operational conditions and predicts the amount of energy consumed in different conditions based on the real world data. The proposed machine learning based prediction and optimization system in this project is based on this conceptual framework.

A lot of preprocessing was done to address the missing and the exceptionally large values, normalized the numbers attributes as well as coded the categorical values. Predictive performance was improved by applying advanced feature engineering methods. These were time based characteristics, lag variables, rolling statistics and smoothing signals through filtering methods and periodic encoding to detect cyclical characteristics in network activity.

A number of machine learning models were experimented and compared. These included:

- TabTransformer for handling categorical feature interactions
- TabNet combined with XGBoost for attention-based learning on tabular data
- Deep neural network models implemented using Keras
- FastAI tabular learners
- Gradient boosting methods such as XGBoost

Although these models showed competitive output, the Random Forest Regressor consistently showed better output in terms of prediction accuracy, robustness, and time stability in terms of

results during the time-series validation. This capability to support non-linear relationships, mixed types of features and engineered time-related features made it especially well-suited in this purpose.

The last model was found to be having a high predictive performance and a high degree of generalization, which makes it reliable in predicting hourly base station energy. Based on these forecasts, it is possible to suggest such intelligent energy optimization techniques as dynamic power modulation and adaptive configuration control.

### **1.2.5 Scope of the Project**

This project covers the following scope:

- The 5G base station hourly energy consumption predictive modeling.
- Using machine learning to process structured telecom data.
- Determination of vital energy-driving forces.
- Prototyping of energy prediction system.

The project is not real-time deployment in an actual live telecom infrastructure but is designed as a research level outlay that can be furthered in the industry to be applied to other industries. Future developments can involve real time adaptive optimization, network management system interoperability and support to 6G conditions.

### **1.2.6 Organization of the Report**

The rest of this report will be structured as follows:

- Chapter 2 – Literature Review: Reviews the current literature on energy efficiency wireless networks and machine learning application in telecom energy management.
- Chapter 3 – Proposed Methodology: This section will outline the dataset, preprocessing techniques, feature engineering methods, model development and system architecture.
- Chapter 4 – Results and Discussion: The chapter gives results of the experiment, metrics of evaluation, comparison of the performance with the predictive models analysis.
- Chapter 5 – Conclusion and Future Scope: Provides a concluding statement of the gains of the project, limitations, and future research opportunities.

# CHAPTER 2

## LITERATURE REVIEW

### 2.1 Overview

Telecommunications in the modern world has been altered owing to the rapid deployment of fifth-generation (5G) wireless networks with high data speeds, low-latency communication, and high levels of connectivity. However, besides these technological benefits, there is a major problem that is energy consumption. A study by organizations such as the International Telecommunication Union indicates that the device setup, the level of deployment density, and the amount of traffic placed on it have a significant influence on the amount of energy consumed by the 5G infrastructure.

Most of the energy involved in the Radio Access Network (RAN) goes to the base stations which, in most cases, has been referred to in the literature as the biggest share of energy in the network. The optimization of power consumption is important to sustainable telecommunications with network operators deploying elaborate antenna designs and dense cell patterns.

This chapter analyzes previous studies on:

- Energy consumption behaviour in 5G networks.
- The comparison with previous generations (4G/LTE) is unlike.
- Base station factors that influence the amount of power consumed.
- Algorithms of modeling and machine learning.
- Green networks utility deployment techniques.

## 2.2 Energy Consumption Characteristics of 5 G Networks.

Research has it that 5G energy consumption is dynamic rather than being static. Unlike legacy networks, 5G base stations reconfigure their power consumption dynamically based on signal processing load, antenna operation, and load.

The last studies revealed that:

- The higher the user load, the higher the amount of energy consumed.
- Still, idle power consumption is high due to signaling.
- Hardware configuration has a great influence on efficiency.
- Its technical publications claim that efficient energy management practices are necessary since power use does not reduce in line with reduction of traffic.

## 2.3 Key Characteristics of 5G Energy Consumption

TABLE 2.1 Factors Affecting Energy Consumption

| Factor             | Effect On Power Usage                                  |
|--------------------|--------------------------------------------------------|
| Traffic Load       | Higher Throughput Increasing Processing Energy         |
| Antenna Count      | Massive MIMO increases hardware energy demand          |
| Frequency Bands    | High-frequency operation raises signal processing cost |
| Deployment Density | More base stations increase network energy footprint   |

## 2.4 Comparing the Energy Efficiency of 4G and 5G

There exists an enormous literature comparing current 5G implementations and LTE (4G) networks. The more complex equipment introduced by 5G is represented by massive MIMO arrays and beamforming radios, but it also uses far less energy per transmitted bit.

However, the general energy consumption is generally increased due to:

- A rise in the need for data
- High concentrations of tiny cells.
- Complexity of processing increases.

Consequently, despite the technical efficiency of 5G, networks are transmitting a lot more data, and this increases the total energy usage.

## 2.5 Energy Comparison of Network Generations

TABLE 2.2 4G VS 5G

| Feature             | 4G (LTE) | 5G     |
|---------------------|----------|--------|
| Energy Per Bit      | Higher   | Lower  |
| Hardware Complexity | Moderate | Hard   |
| Network Density     | Low      | High   |
| Total Energy Usage  | Lower    | Higher |

## **2.6 Elements Affecting Base Station Power Usage**

Many important factors that affect base station energy consumption are commonly found in research:

- Multidimensional signal processing.
- The number of used antennas.
- Demand and traffic of the user.
- Parameters of radio setup

During periods of idleness, base stations are rather energy-consuming since the control signaling and synchronization should not be turned off. Thus, the standards bodies like the 3rd Generation Partnership Project have suggested energy-saving features like using micro-sleep modes and antenna muting.

## **2.7 Modeling Techniques for Energy Efficiency**

Modeling power consumption is one of the most important steps in the optimization process. Earlier work has suggested both the use of analytical and machine learning-based approaches.

Models of Analysis

- The traditional models estimate energy by the following:
- Levels of throughput
- Power transmission
- Antenna configuration.

Even though the results obtained by them can be interpreted, they struggle to discover nonlinear patterns.

According to recent studies, the complex relationship between the network parameters and energy consumption is recapturable through machine learning algorithms.



The characteristics used in these models include:

- Patterns of traffic load
- Use by time of day
- Configuration of the base station
- Environmental states

## 2.8 Modeling Techniques in Literature

TABLE 2.3 Limitations And Advantages Of Different Models

| Model Type           | Limitations              | Advantage                      |
|----------------------|--------------------------|--------------------------------|
| Analytical Models    | Interpretable, simple    | Poor At Non-Linear Predictions |
| Regression Models    | Moderate Accuracy        | Sensitive To Feature Scaling   |
| Tree-based Ensembles | High accuracy            | Less interpretable             |
| Deep Learning Models | Capture complex patterns | Requires Large Datasets        |

## 2.9 Energy Forecasting using Machine Learning

The use of machine learning to predict telecom network is a trend being popular as per the recent research. Random Forest, XGBoost, LightGBM and LSTM net algorithms are commonly evaluated in terms of energy forecasting applications.

Critical conclusions are:

- Feature engineering frequently outperforms complex systems in terms of performance improvement.
- On structured datasets, ensemble approaches consistently yield good results.
- improve prediction quality.

Given that ensemble techniques employing engineered features yielded exceptionally high predicted accuracy, your project is consistent with these findings.

TABLE 2.4 Strengths Of Different Models

| Model             | Typical Strength                       |
|-------------------|----------------------------------------|
| Random Forest     | Robust and stable predictions          |
| Gradient Boosting | High accuracy with tuned parameters    |
| LSTM Networks     | Effective for temporal dependencies    |
| Hybrid Models     | Combine temporal and tabular strengths |

## 2.10 Green 5G Network Deployment Techniques

Several tactics are suggested by researchers to lower network energy consumption:

- Switching between base stations dynamically
- Sleep modes that consider traffic
- AI-powered load distribution
- Choosing gear that uses less energy

Combining these strategies can dramatically lower network power consumption without sacrificing service quality, according to simulation studies and field observations.

## 2.11 Patterns of Energy Use in Urban and Rural Areas

Energy use varies significantly by geographic environment, according to field measurements:

- Because of the increased traffic density, urban stations need more energy. Underutilization causes energy waste at rural stations.
- Therefore, compared to uniform models, location-aware models offer more accurate energy projections.

### 2.11.1 Urban vs Rural Consumption Trends

TABLE 2.5 Urban Vs Rural Consumption Comparison

| Environment | Key Challenge    | Optimization Approach       |
|-------------|------------------|-----------------------------|
| Urban Areas | High load        | Traffic Adaptive Scheduling |
| Rural Areas | Underutilization | Dynamic power scaling       |

## 2.12 Identification of Research Gaps

Several gaps are revealed by the reviewed literature:

- Lack of generalized models across multiple base station types
- Absence of generalized models for many types of base stations
- Predictive modeling and feature engineering are not fully integrated.
- Real-time, flexible forecasting systems are required.
- Inadequate comparison between ML and DL methodologies

By creating feature-rich datasets and methodically assessing several model classes, your project fills up these gaps.

## **2.13 Literature Review Synopsis**

Combined, it can be seen that the reviewed literature demonstrates that 5G network energy efficiency is a complicated problem that depends on traffic dynamics, the network configuration, hardware design, and intelligent optimization approaches.

ML-based models have shown much promise with regard to making the appropriate prediction of the energy consumption, placed in the context of using ensemble methods and designed traits. The studies that should be carried out in the future should however focus on the realtime adaptive systems which are capable of generalizing themselves in different deployment environments.

The theoretical basis of the proposed research is built in this chapter, as well as the importance of predictive modeling in making the next-generation networks energy-saving and sustainable.

# **CHAPTER 3**

## **PROPOSED METHODOLOGY**

### **3.1 Overview of System Implementation**

The proposed system implementation step is aimed at the creation of an efficient machine learning pipeline of 5G base stations energy consumption prediction and optimization based on engineered telecom data.

The system combines data cleaning, feature engineering, benchmarking of models, ensemble learning, and final models deployment in a workflow. This project involves focusing on the time-series feature engineering, signal processing method, and the sophisticated ensemble modeling in order to correctly predict the energy usage patterns at various base stations and in various conditions over time, as compared to the traditional prediction systems.

It has been implemented with Python-based machine learning frameworks and run in a controlled experimental setting in order to guarantee reproducibility, scalability, and model reliability.

### **3.2 Development Environment and Tools**

#### **3.2.1 Programming Language**

Python is a machine learning and data science ecosystem that was chosen strategically. It was based on its high readability and excellent libraries such as scikit-learn, pandas and NumPy.

Development of specialized systems such as TensorFlow and PyTorch facilitated the use of deep learning. Such an integrated environment simplified the development, reduced the integration problems and used the support of the community. Python therefore provided an effective and current system construction.

TABLE 3.1. Libraries and Frameworks Used

| Category              | Tools Used                                      | Purpose                          |
|-----------------------|-------------------------------------------------|----------------------------------|
| Data Processing       | Pandas, NumPy                                   | Data cleaning and transformation |
| Visualization         | Seaborn, Matplotlib                             | Exploratory analysis             |
| Machine Learning      | Scikit-learn, XGBoost, FastAI, TensorFlow/Keras | Model training and evaluation    |
| Signal Processing     | SciPy                                           | Feature smoothing and filtering  |
| Deep Learning Backend | PyTorch (via FastAI)                            | Tabular neural modeling          |
| Validation            | Scikit-learn (KFold, Time Series Split)         | Model validation                 |

### 3.2.2 Execution Platform

The project was developed/executed using the Kaggle Notebook Environment and with the support of a GPU/CPU Hybrid Execution setup as it is a more powerful hardware, which thus greatly supported the machine learning models training process and the evaluation process of the models as well. This was a high-performance environment that was essential in managing the huge amount of computation and delivery of timely results. The data set contained more than 160 000 engineered samples, with a greater number of than 90 engineered features.

### 3.3 Dataset Acquisition and Integration

#### 3.3.1 Dataset Sources

The data set utilized in this project has a variety of telecom data which are Base Station (BS) Data, Cell Load Data, Energy Consumption Logs and Network Configuration Data. The merging of these datasets was done according to similar keys like Time, Base Station ID (BS), and Cell Name. This merging has made it possible to produce a single rich feature data which can be used in model in machine learning.

TABLE 3.2. Dataset Characteristics

| Parameter                    | Value                    |
|------------------------------|--------------------------|
| Total Samples                | 160,000+                 |
| Features (After Engineering) | 90+                      |
| Target Variable              | Energy Consumption       |
| Data Type                    | Time-Series Telecom Data |

### 3.4 Data Preprocessing Pipeline

#### 3.4.1 Handling Missing and Inconsistent Values

In preprocessing, forward fill, interpolation and controlled imputation methods were used in handling missing values that occurred during lag and rolling feature generation processes. This methodological practice played a very important role in ensuring the integrity of data and making the following machine learning models to have as much predictive power as possible. Values that were extreme and infinite were identified and replaced to bring about numerical stability in model training.

### **3.4.2 Data Cleaning and Transformation**

Preprocessing steps as follows were important:

- Elimination of superfluous columns (Time, CellName, etc.).
- Transformation of categorical telecom characteristics (RUType, Mode).
- Numerical variable normalization.
- Lag features create NaN values which need to be dealt with.
- Ranking data in time-series integrity.

This preprocessing helped a lot in enhancing the stability of models and minimizing noise during training.

## **3.5 Advanced Feature Engineering Implementation**

The main asset of this project was feature engineering and it made this project to have a high predictive accuracy level particularly by converting raw data into meaningful inputs which enhanced the model performance and generalization significantly.

### **3.5.1 Time-Based Feature Extraction**

The subsequent features were obtained to identify the time variations in energy use: Hour of Day, Day of Month, Weekday Number as well as Cyclical Hour Encoding based on Spline Transformer. The cyclical spline transformation was particularly used to conserve the natural periodic activity which can be witnessed in the energy use patterns on a daily basis, which is essential in proper forecasting and modeling.

### **3.5.2 Lag Features (Temporal Memory Modeling)**

Lag features were created to capture the past history behaviour into the prediction model: Energy Lag (1 hour, 3 hours, 6 hours, 24 hours), Load Lag Features and ESMODE Historical Values. Such characteristics as Energy\_lag\_1 (Energy at t-1) and Energy\_lag\_24 (Energy at t-24) allow the model to acquire temporal correlations and repetitive consumption patterns.



### **3.5.3 Rolling Statistical Features**

To evenly spread out noise and extract patterns of trend, rolling window statistics were calculated, and Rolling Mean Energy, Rolling Maximum Energy, Rolling Minimum Load and Rolling Window statistics were calculated over 3, 6, 12 and 24 hours, all good exercise that made the model robust to short-term fluctuations.

### **3.5.4 Signal Processing Features**

In order to sharpen telecom load signals, state-of-the-art filtering was used:

- Savitzky-Golay Filter (Load Smoothing & Derivatives)
- Butterworth Filter (Noise Reduction)
- Load Gradient and Differential Features.

The methods aided in deriving concealed designs of noisy telecommunications signals.

### **3.5.5 Interaction and Derived Features**

Further domain specific characteristics were developed to enhance the model learning performance such as Load  $\times$  Antennas Interaction, Energy Efficiency Metrics, Load Change Rate, and Load Binning to represent loads as categorical values and this multi-level feature engineering gave great gains.

## **3.6 Model Development and Benchmarking Strategy**

Various machine learning and deep learning models including the different neural network structures and classical algorithms were constructed and later evaluated rigorously. Such extensive measure was to ensure the most suitable predictive model in this complicated classification process was identified and picked according to performance measures.

### **3.6.1 Deep Learning Models Tested**

The TabTransformer, FastAI Tabular Neural Network, and the Keras Multi-Layer Perceptron (MLP) were the models employed in order to model the nonlinear relationships in the tabular telecom data.

### **3.6.2 Ensemble Models Implemented**

Enhancing the predictive accuracy substantially required an in-depth search in diverse ensemble models where single-model structures were no longer used. In particular, three different ensemble strategies were explored, including, first, a hybrid approach based on TabNet learning of features and XGBoost strong predictors on the output of the latter; second, an advanced one, where Keras deep learning model output is used as the input to an XGBoost meta-learner; and third, a smart combination of FastAI models and a Keras-written model. The essence was that the effective capturing of the features of neural models is to be joyfully combined with a powerful generalization and performance of tree-based algorithms such as XGBoost. Such a combined strategy eventually brought a significant increase in predictive accuracy.

## **3.7 Final Model Selection: Random Forest**

Random Forest has been chosen as the final model to be used following several experiments because it performs better when using engineered tabular data.

Reasons for Selection:

- Manages nonlinear telecommunication information.
- Robust against overfitting
- Exceptional use of lag and roll performance.
- Large interpretability through feature significance.
- Constant performance over time-series validation.

The model had almost perfect predictive ability because of solid structured associations in engineered features.

### **3.8 Training and Validation Methodology**

#### **3.8.1 Cross-Validation Strategy**

A broad range of validation methods was used to strictly determine the performance of the model and make sure the model is capable of a sound generalization. It involved K-Fold Cross Validation, in this case, 5-fold, in order to assess the uniformity of the model in varying processes of the information. Moreover, since the data in it is time-based, Time-Series Cross Validation was carried out to ensure the chronological order of training and testing. Lastly, testing apparently an out of distribution test which is known as Distant Future Testing was done to ensure that indeed the model was not given a one-sided evaluation as to whether it did not overfit the data to certain future temporal data with no prior data as it was originally supposed to do.

### **3.9 System Architecture and Workflow**

The entire system workflow is made up of the following phases:

1. Data Collection and Merging
2. Data Preprocessing
3. Feature Engineering
4. Model Training
5. Model Evaluation
6. Prediction Generation
7. Output Export (CSV)

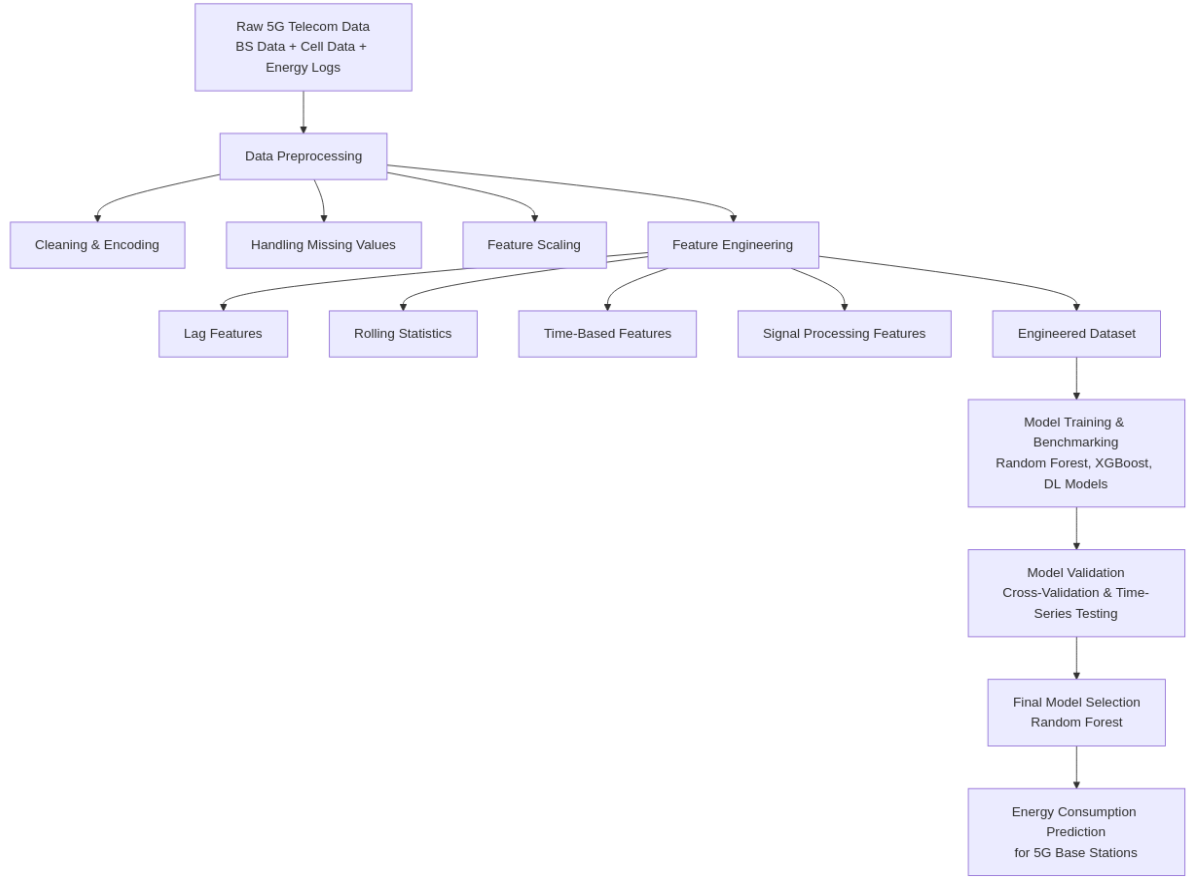


FIGURE 3.1. System Architecture

### 3.10 Implementation Challenges and Solutions

There are various obstacles that were experienced during the project. These were controlling the risk of NaN explosion in the generation of lag features, controlling the risk of data leakages that occurred due to time-series modeling, managing the issue of model instability that occurred due the utilization of crude preprocessing methods, and effectively operating at the high-dimensional feature space.

The measures that were taken comprised controlled imputation plans, time-sensitive dataset separation, validation of feature significance, and powerful group benchmarking. These methodological enhancements proved very very instrumental towards not only improving the predictive quality of the end model, but also improving its interpretability.

### **3.11 Output Generation and Deployment**

The last step was to create Predicted Energy Consumption based on the trained model on unknown data. The findings, such as the Base Station ID and the prediction are sent out as an easily digestible CSV file. Importantly, the scaler, the data on feature importance and the trained model files were all saved. This guarantees auditory reproducibility and eases further implementation into production or real-time systems.

# **CHAPTER 4**

## **RESULTS AND DISCUSSION**

### **4.1 Introduction**

The chapter provides the experimental findings of the proposed energy consumption prediction framework of 5G base stations. The aim of the experiments was to compare various machine learning and deep learning methods, learn the effect of feature engineering, and determine which model was the most reliable to apply in practice.

They did a systematic evaluation procedure which involved model comparison, feature importance analysis, time-series validation and testing on unseen future data. The discussion is aimed at learning model behaviour in verbal form instead of documenting numerical performance only.

### **4.2 Experimental Setup**

The experiments were conducted on a combined set of hourly base station records (network load, configuration parameters, energy-saving modes, and measured energy consumption) of the merging dataset. The resulting dataset, which was produced after preprocessing and feature engineering, comprised around 160,000 records, and it had almost 90 informative features.

### 4.2.1 Feature Engineering

In order to generate temporal and operational behaviour of base stations, the following feature engineering strategies were introduced:

- Mining of time related features like hour, day and weekday.
- Periodic cyclical time spline transformations.
- Lag characteristic of historical values of energy and load.
- Sliding statistical characteristics (mean, max, min).
- Signal smoothing with Butterworth and Savitzky Golay filter.
- Combination characteristics of interaction parameters inserting load and hardware parameters.
- Elimination of low-value and high-sparsity features.

These artificial features contributed a lot in enhancing the capability of the model to generate temporal dependencies.

### 4.3 Evaluation Metrics

Given that the issue is a regression task, the evaluation of performance was made through:

- **Mean Absolute Error (MAE)** – mean true value versus predicted value.
- **Root Mean Squared Error (RMSE)** – puts more weight on larger prediction errors.
- **Mean Absolute Percentage Error (MAPE)** – relative measure of error.
- **Coefficient of Determination ( $R^2$ )** – explicates variance included by the model.

Also, time-series cross-validation and distant-future testing were conducted as an embodiment of robustness and to ensure against temporal leakage.

## 4.4 Model Comparison

### 4.4.1 Deep Learning and Hybrid Models

First a few sophisticated models were tested:

TABLE 4.1. Model Performance Comparison.

| Model            | MAE    | MAPE   | Accuracy (Approx.) |
|------------------|--------|--------|--------------------|
| TabTransformer   | 3.3257 | 0.1289 | 87.11%             |
| TabNet + XGBoost | 2.7663 | 0.1074 | 89.26%             |
| Keras + XGBoost  | 2.6161 | 0.1081 | 89.19%             |
| FastAI + Keras   | 3.3307 | 0.1352 | 86.48%             |

These models are not very sensitive to preprocessing and hyperparameter settings even though they captured nonlinear relationships.



#### 4.4.2 Traditional Machine Learning Models

Classical machine learning models were benchmarked following the enhancement of feature engineering.

TABLE 4.2.Evaluation Metrics of Traditional Machine Learning Models.

| Model             | Test RMSE     | Test MAE      | Test R <sup>2</sup> |
|-------------------|---------------|---------------|---------------------|
| Random Forest     | <b>0.4853</b> | <b>0.1119</b> | <b>0.9990</b>       |
| Gradient Boosting | 0.5215        | 0.2843        | 0.9988              |
| XGBoost           | 0.5288        | 0.2905        | 0.9988              |
| LightGBM          | 0.5823        | 0.2965        | 0.9985              |
| Linear Regression | 2.0389        | 1.1211        | 0.9820              |

Random Forest was the highest performer with high predictive stability and minimum error.

#### 4.5 Random Forest Model Analysis

Random Forest is an ensemble method that utilizes multiple decision trees, and works by building multiple decision trees using randomly sampled data. The trees would learn their own relationships and the end predictions would be obtained by averaging the outputs of the trees. This avoids the overfitting characteristic of a single decision tree that it tends to have, and results in more stable results and improved generalization.

This randomness is done through bootstrap aggregating (bagging) to choose data, and at every split random feature selection. Bagging uses a wholly random sample of training data. This is so that the trees are trained using separate datasets. Random Forest uses only a random sample of features in case of a tree being constructed. This correlates the trees, and this makes the model very strong. Majority voting is the last classification prediction; and it is average individual tree predictions in case of regression. Random Forest is a very popular algorithm,

which is due to its effectiveness.

The model was successful since:

- It was naturally capable of working with a variety of features.
- The use of random feature selection eliminated correlation of models.
- Prediction averaging reduced overfitting.
- The nonlinear elements between load, time and configuration parameters were well captured.

Random Forest became very appropriate with complex energy consumption trends.

## 4.6 Feature Importance Analysis

The importance of features analysis revealed that historical energy features were the most important predictors.

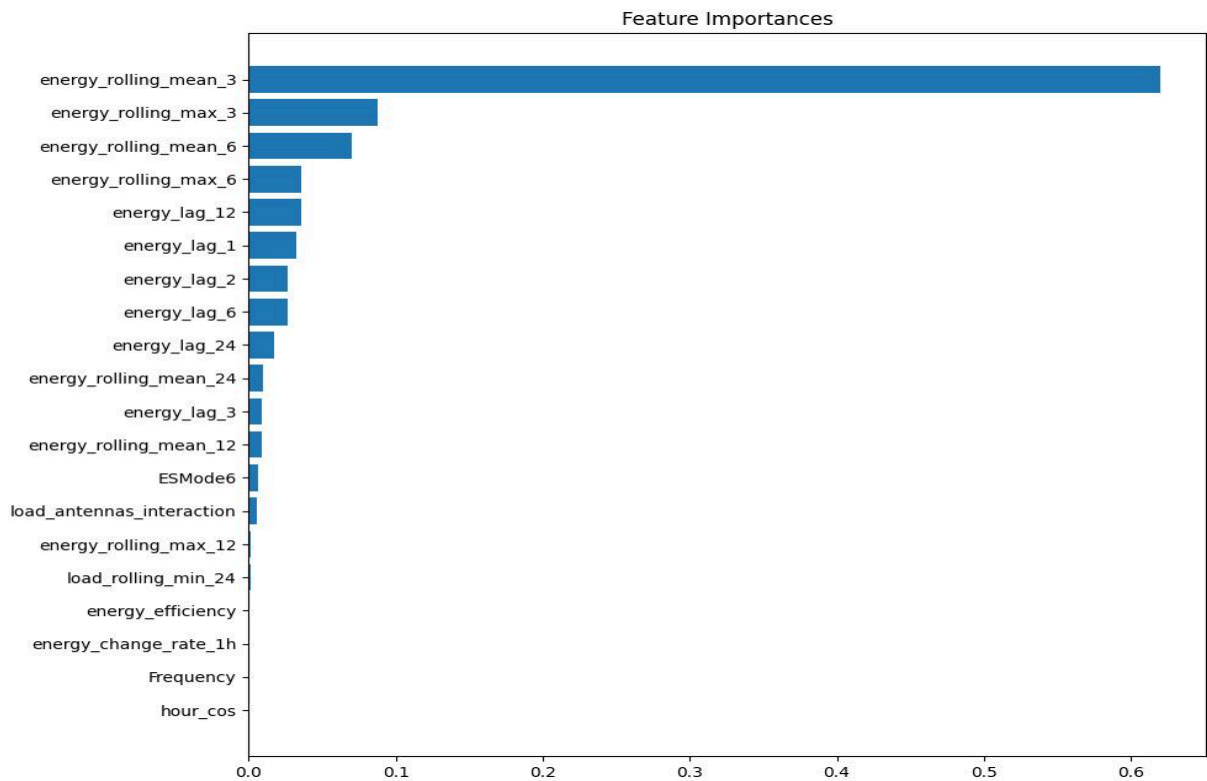


Figure 4.1. Feature Importance Plot.

The most donating features were:

- energy\_rolling\_mean\_3
- energy\_rolling\_max\_3
- energy\_lag\_1, energy\_lag\_2, energy\_lag\_6
- energy\_rolling\_mean\_6
- load interaction features

These characteristics prevailing prove that short-term temporal memory is one of the key factors that determine energy consumption.

## **4.7 Validation and Generalization Testing**

### **4.7.1 Feature Dependency Testing**

To confirm the reliability of the models, experiments were conducted with and without features derived by energy.

Findings demonstrated that a significant decrease in performance was observed when historical energy characteristics were eliminated, meaning that the model does not memorize information, but learns useful temporal-dependency relationships.

### **4.7.2 Time-Series Cross Validation**

Cross-validation was found to be time-aware:

- Average  $R^2 \approx 0.9849$
- Low variance across folds

This means that there are regular performances across time intervals.

### 4.7.3 Distant Future Testing

The model was trained using the past time windows and assessed using future unknown time.

Results:

- Train  $R^2 \approx 0.9992$
- Test  $R^2 \approx 0.9952$

These findings show high generalization ability to make predictions on the real world.

## 4.8 Actual vs Predicted Energy Analysis

### 4.8.1 Scatter Plot Analysis

This visual evaluation of the model performance is through the scatter plot comparing actual and the predicted energy values. Preferably, predictions are supposed to be on the diagonal reference line, which is perfect prediction.

The results are plotted and the result indicates a strong alignment along the diagonal line, which implies that the model prediction was in agreement with the actual measurements of energy.

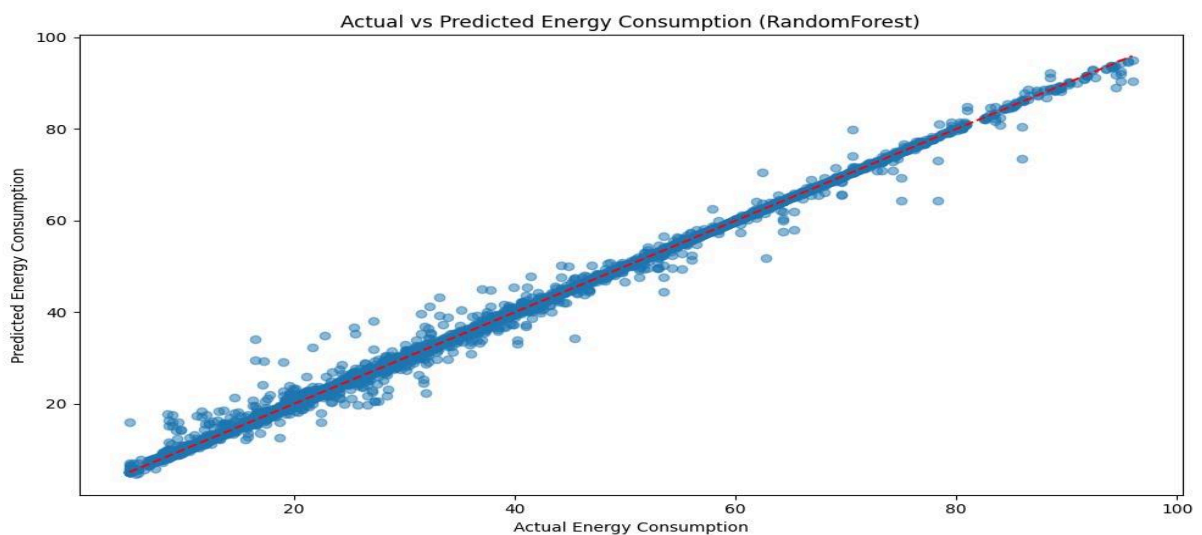


Figure 4.2. Actual vs Predicted Energy Consumption (Scatter Plot).

Key observations include:

- The forecasts are close to the optimal diagonal line.
- There is low error distribution in the various energy ranges.
- The number of outliers is very few.

These results affirm that the random forest model is a successful model to represent underlying energy consumption trends.

#### 4.8.2 Time-Series Prediction Analysis

Although the scatter plot can be used to judge overall accuracy, it is also possible to analyse predictions with time to determine the stability and predictability of the model in time.

Time-series comparison reveals that forecasted values are much closer to the actual energy trends over the course of the test, which means that the model can trace the fluctuations in the time.

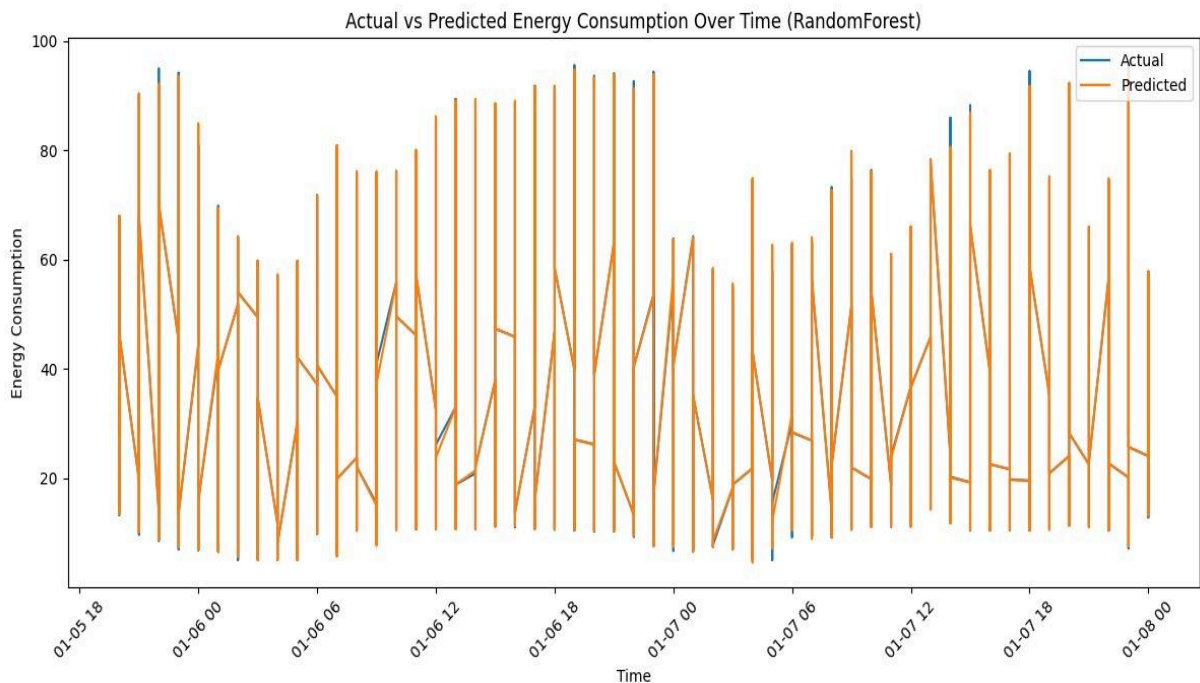


Figure 4.3. Actual vs Predicted Energy Consumption Over Time.

The fact that the actual and predicted curves are too close to each other implies:

- Effective temporal generalization.
- Constant operation on different loads.
- Minimal delay in response time in prediction.

### 4.8.3 Feature Relationship Analysis

The ability to know the correlation between engineered features is useful in understanding the behavior of a model and contribution by features.

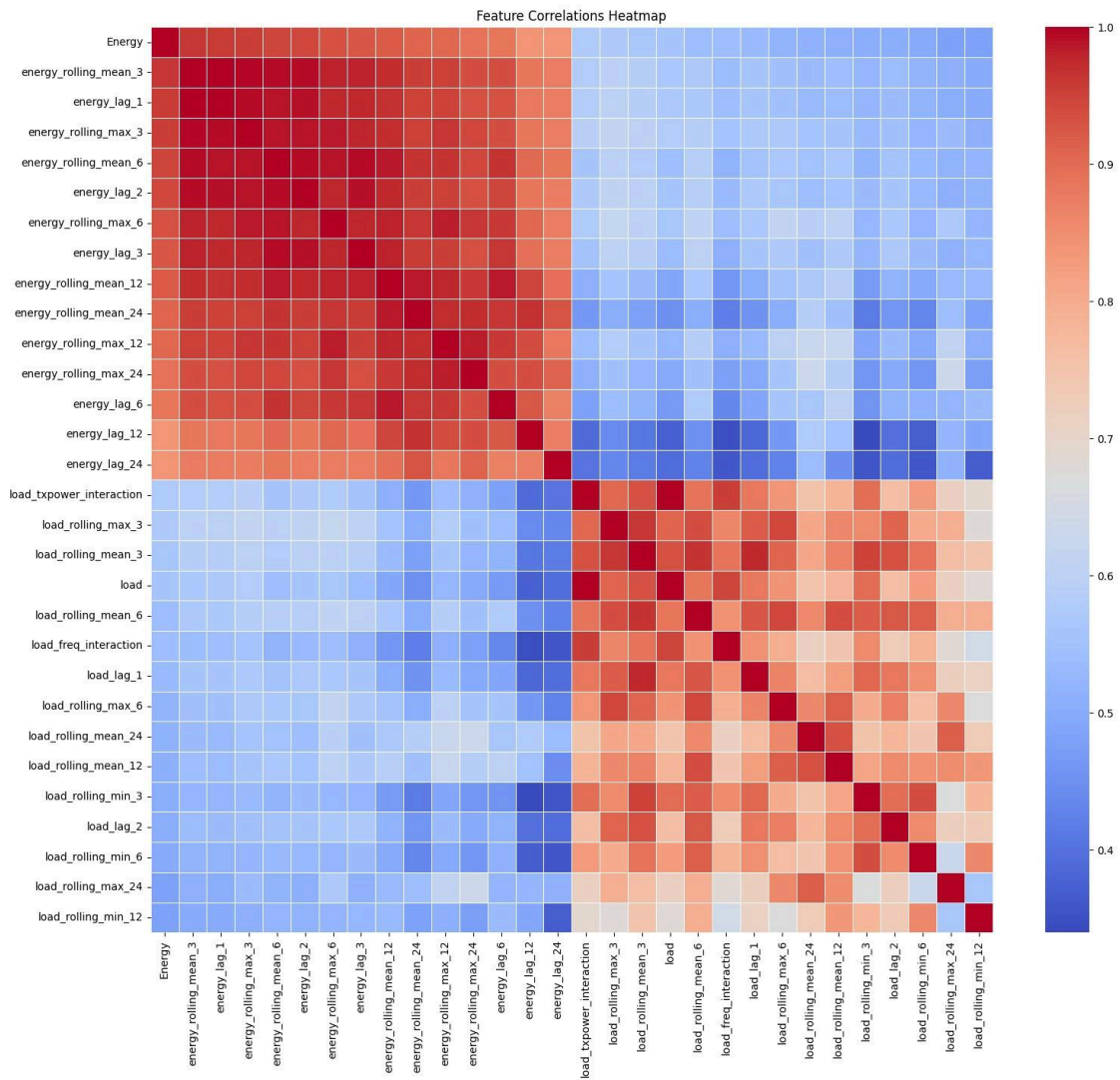


Figure 4.4. Feature Correlation Heatmap.

The heatmap shows associations between time and energy-based features, which allow considering feature engineering as the key to the enhancement of prediction performance.

## **4.9 Discussion**

Based on the experiments, a number of insights have been realized:

1. The model complexity had a lesser contribution to the performance improvement when compared to feature engineering.
2. Temporal and rolling characteristics showed a great deal of improvement in predicting.
3. The ensemble tree-based models performed better when compared to the deep learning models in this tabular data.
4. Random Forest was accurate and its interpretation and stability were high.

The presented framework, on the whole, is able to model the trends in energy consumption in 5G base stations and is highly applicable in practice.

# **CHAPTER 5**

## **CONCLUSION AND FUTURE SCOPE**

### **5.1 Conclusion**

In this project, they provided an entire machine learning architecture in predicting energy use in 5G base stations. The experiment was a combination of domain knowledge, widespread feature engineering, and model benchmarking to develop a predictive system which was reliable.

Its significant accomplishments are:

- Merging and pre processing several network data sets.
- Construction of time and statistical characteristics which mirror energy behaviour.
- Comparison of deep and classic machine learning models.
- Strict validation in terms of time-series cross-validation and future period testing.

Random Forest provided the highest accuracy and good generalization on unseen data when compared to other models, and thus it was the best among all models.

The paper establishes that energy in contemporary cellular networks is exceptionally reliant on temporary dynamics and interaction of network loads.



## 5.2 Contributions of the Work

The main deliverables of this project are:

- End to end energy prediction pipeline development.
- Evidence of the significance of feature engineering when tackling a tabular time-series problem.
- Checking of strong model performance without having to use excessively complex architecture.
- Distribution of a scalable structure that can be used in sustainable energy management.

## 5.3 Future Scope

This work can be further improved by a number of directions:

### 1. Deep Learning Models in sequence

Application of LSTM or GRU models to longer temporal dependencies.

### 2. Advanced Ensemble Stacking

Meta-learning with Random Forest, XGBoost and LightGBM.

### 3. Real-Time Deployment

Connection to network monitoring systems of live energy forecasting.

### 4. Feature Optimization

Eliminating unnecessary complexity of models by choosing the best ranking features to infer with.

### 5. Adaptive Energy Control

Predictions can be used to automatically optimise energy saving modes in live networks.

## **5.4 Final Remarks**

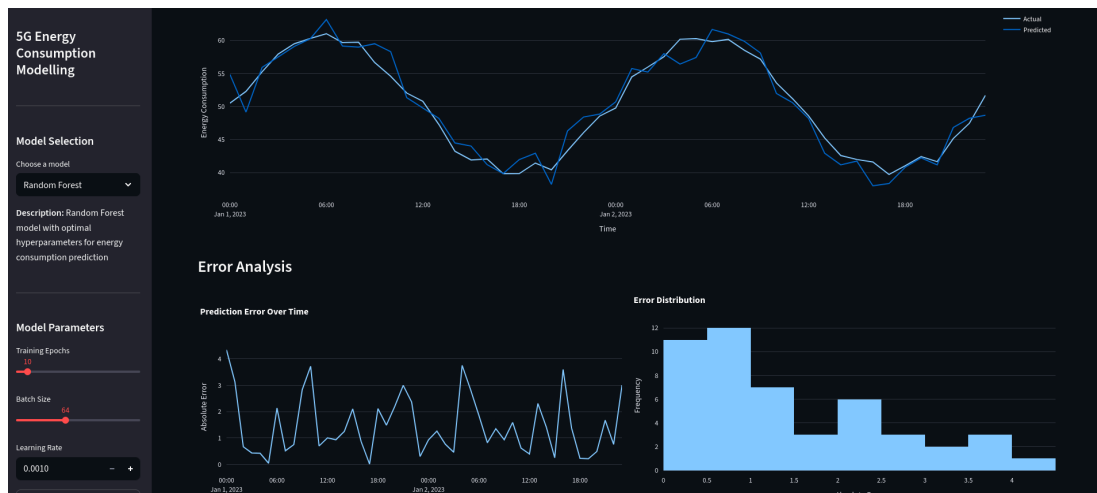
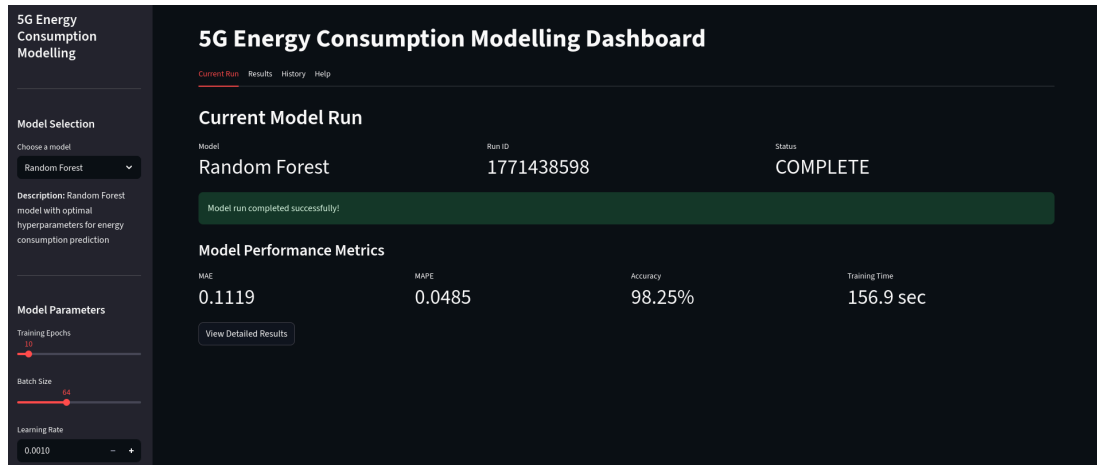
This project shows that to construct efficient machine learning systems, one needs to be careful in data preparation, meaningful feature engineering, and systematic validation. The model developed is not only well-performing in terms of prediction but also interpretable and scalable, which makes it an appropriate choice in future studies as well as application in the sustainable 5G network management.

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# APPENDIX 1

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# RESEARCH PAPER

## Sustainable Energy Management in 5G Networks

Publisher: IEEE

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3

Full

Text Views



### Abstract

Document Sections

I. Introduction

II. Experimental Setup

III. Feature Engineering and  
Model Selection

IV. Suitability of Random  
Forest for Handling  
Complex 5G Energy Data

V. Result and Discussion

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### Abstract:

With the rollout of 5G networks, energy consumption has become a pressing concern, even though 5G is generally more efficient than 4G. A significant share of this consumption—over 70 %—comes from the radio access network (RAN), particularly the base stations. To tackle this issue, it is essential to fine-tune base station configurations and apply effective energy-saving strategies. Factors such as hardware design, configuration settings, network traffic, and applied optimization techniques play a critical role in determining energy usage. Building reliable models that can accurately predict energy consumption is therefore key to achieving sustainable and efficient 5G deployments. This study further explores how machine learning can be leveraged to address these challenges and provide data-driven solutions for enhancing network energy efficiency.

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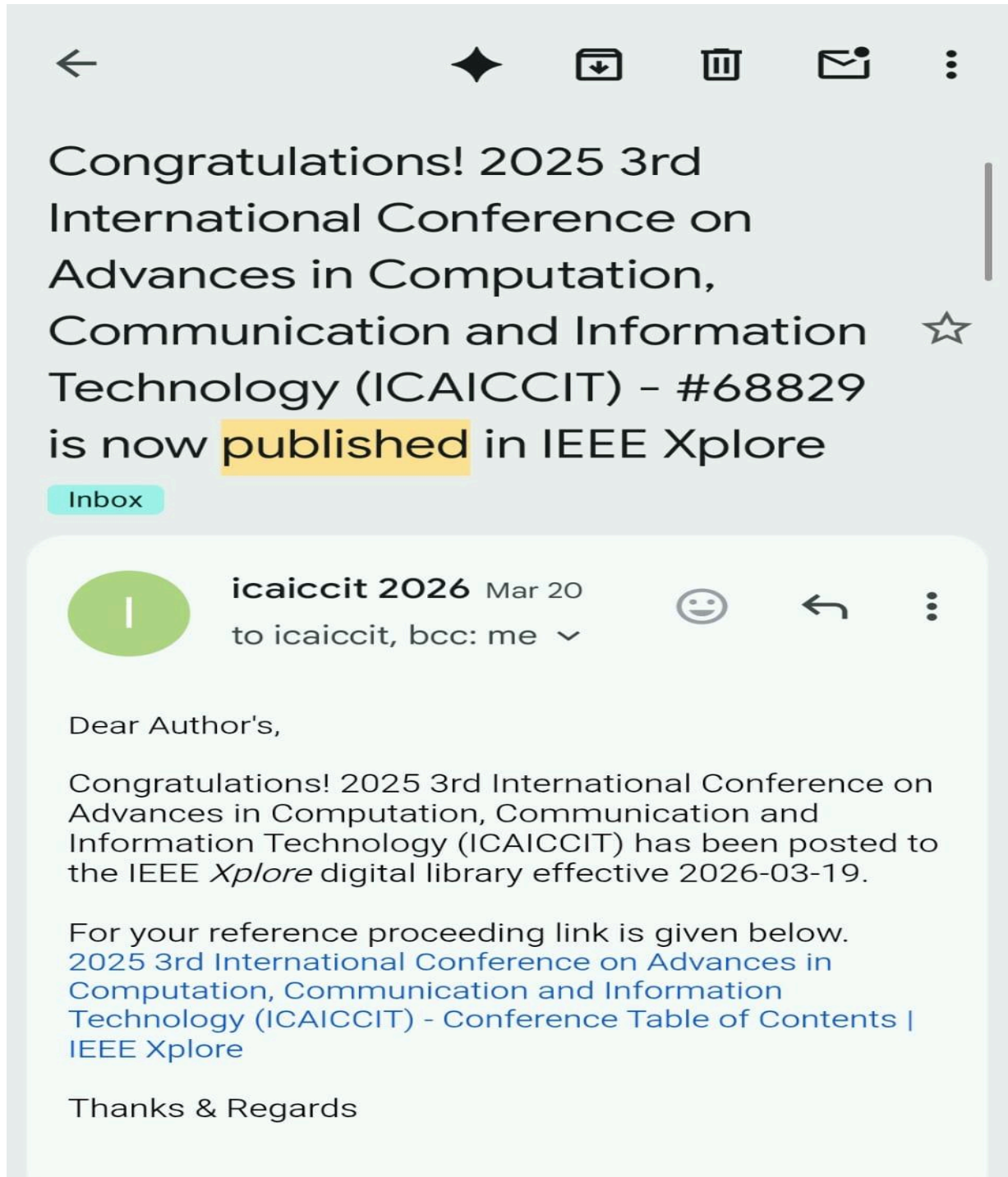
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# PLAGIARISM REPORT

## Sustainable Energy Management in 5G Networks

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**Abstract**—With the rollout of 5G networks, energy consumption has become a pressing concern, even though 5G is generally more efficient than 4G. A significant share of this consumption—over 70%—comes from the radio access network (RAN), particularly the base stations. To tackle this issue, it is essential to fine-tune base station configurations and apply effective energy-saving strategies. Factors such as hardware design, configuration settings, network traffic, and applied optimization techniques play a critical role in determining energy usage. Building reliable models that can accurately predict energy consumption is therefore key to achieving sustainable and efficient 5G deployments. This study further explores how machine learning can be leveraged to address these challenges and provide data-driven solutions for enhancing network energy efficiency.

### I. INTRODUCTION

Today is the world of 5G network as it provides exceptionally high data speed, minimal latency, and the ability to connect to the vast network of smart devices and IoT applications. Due to this, everyone wants to switch over to it, as it makes the path for their endless success and provides ease in their day-to-day life.

Although 5G networks are almost four times more energy-efficient than the previous ones, they also have certain side effects, including higher energy consumption. Various factors such as the requirement for denser networks, deployment at higher frequencies, wider bandwidth, and the processing needed for support contribute to this increased energy demand.

The major portion of energy consumption originates from mobile networks, which is largely attributed to the Radio Access Networks (RAN), where base stations serve as the primary contributors.

On the other hand, factors such as data centers and transport systems have a negligible influence on the overall energy consumption of 5G networks.

All these inefficiencies demand an effective solution that can overcome the imbalance by focusing on optimizing base station configurations along with implementing efficient energy-conservation techniques. It is essential to develop a comprehensive model that can accurately forecast the interdependence of various factors on power usage, thereby enabling the attainment of higher overall efficiency in 5G networks.

This paper discusses the challenges and possible solutions related to the issue of energy consumption in 5G networks. Key aspects such as base station architecture, configuration methods, traffic patterns, and energy-conservation mechanisms are analyzed to understand their influence on overall efficiency. To address these concerns and make 5G systems more adaptable, we put forward the use of advanced computational models that are capable of generating predictions, estimating various influencing factors, and identifying the dependencies within 5G networks, with particular reference to algorithms such as random forest.

Our vision is not only focused on developing accurate predictive models but also on transforming diverse scenarios into a unified and consistent framework. In this study, we have primarily assessed performance through metrics such as the weighted mean absolute error and the mean absolute percentage error, which serve as supporting evidence. The central objective of this research is to provide dependable energy estimates while promoting efficient and sustainable deployment of both 4G and 5G networks.

### II. EXPERIMENTAL SETUP

#### A. Base Station Information

This dataset encompasses all the attributes and details related to the base station, which serves as a key element in 5G energy consumption. It includes information such as the cell name, type of radio unit, transmission model, operating frequencies, allocated bandwidth, number of antennas, and the maximum transmission power.

#### B. Cell-Level Statics

This dataset captures the operational activity on an hourly basis. It documents indicators such as cell load, the contribution of specific base stations, cell name, measurement time, and the activation of different energy-saving modes.

#### C. Energy Consumption Records

This dataset focuses on delivering detailed, hour-by-hour information regarding the energy usage of various base stations. Such records play a crucial role in accurately predicting energy



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# Sustainable Energy Management in 5G Networks

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**Abstract**—With the rollout of 5G networks, energy consumption has become a pressing concern, even though 5G is generally more efficient than 4G. A significant share of this consumption—over 70%—comes from the radio access network (RAN), particularly the base stations. To tackle this issue, it is essential to fine-tune base station configurations and apply effective energy-saving strategies. Factors such as hardware design, configuration settings, network traffic, and applied optimization techniques play a critical role in determining energy usage. Building reliable models that can accurately predict energy consumption is therefore key to achieving sustainable and efficient 5G deployments. This study further explores how machine learning can be leveraged to address these challenges and provide data-driven solutions for enhancing network energy efficiency.

**Keywords:** Base Stations, Energy Consumption, Energy Efficiency, 5G Networks, Machine Learning, Random Forest, Sustainable Energy.

## I. INTRODUCTION

Today is the world of 5G network as it provides exceptionally high data speed, minimal latency, and the ability to connect to the vast network of smart devices and IoT applications. Due to this, everyone wants to switch over to it, as it makes the path for their endless success and provides ease in their day-to-day life.

Although 5G networks are almost four times more energy-efficient than the previous ones, they also have certain side effects, including higher energy consumption. Various factors such as the requirement for denser networks, deployment at higher frequencies, wider bandwidth, and the processing needed for support contribute to this increased energy demand. The major portion of energy consumption originates from mobile networks, which is largely attributed to the Radio Access Networks (RAN), where base stations serve as the primary contributors.

On the other hand, factors such as data centers and transport systems have a negligible influence on the overall energy consumption of 5G networks.

All these inefficiencies demand an effective solution that can overcome the imbalance by focusing on optimizing base station configurations along with implementing efficient energy-conservation techniques. It is essential to develop a compre-

hensive model that can accurately forecast the interdependence of various factors on power usage, thereby enabling the attainment of higher overall efficiency in 5G networks.

This paper discusses the challenges and possible solutions related to the issue of energy consumption in 5G networks. Key aspects such as base station architecture, configuration methods, traffic patterns, and energy-conservation mechanisms are analyzed to understand their influence on overall efficiency. To address these concerns and make 5G systems more adaptable, we put forward the use of advanced computational models that are capable of generating predictions, estimating various influencing factors, and identifying the dependencies within 5G networks, with particular reference to algorithms such as random forest.

Our vision is not only focused on developing accurate predictive models but also on transforming diverse scenarios into a unified and consistent framework. In this study, we have primarily assessed performance through metrics such as the weighted mean absolute error and the mean absolute percentage error, which serve as supporting evidence. The central objective of this research is to provide dependable energy estimates while promoting efficient and sustainable deployment of both 4G and 5G networks.

## II. EXPERIMENTAL SETUP

### A. Base Station Information

This dataset encompasses all the attributes and details related to the base station, which serves as a key element in 5G energy consumption. It includes information such as the cell name, type of radio unit, transmission model, operating frequencies, allocated bandwidth, number of antennas, and the maximum transmission power.

### B. Cell-Level Statics

This dataset captures the operational activity on an hourly basis. It documents indicators such as cell load, the contribution of specific base stations, cell name, measurement time, and the activation of different energy-saving modes.

### C. Energy Consumption Records

This dataset focuses on delivering detailed, hour-by-hour information regarding the energy usage of various base stations. Such records play a crucial role in accurately predicting energy consumption patterns under different operational scenarios, thereby highlighting the contribution of each individual base station.

All these datasets collectively provide comprehensive insights into the functioning of 5G networks and reveal the interdependence of multiple factors that contribute to the overall energy consumption.

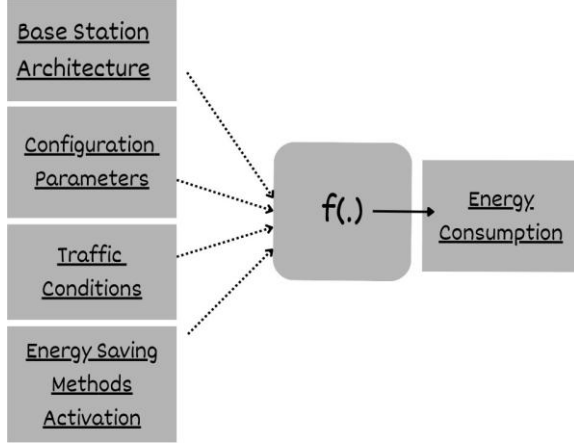


Fig. 1. Factors Affecting Energy Consumptions .

### D. Exploratory Data Analysis and Data Preprocessing

The data collected from different base stations is obtained in raw form, which requires preprocessing and exploration to extract accurate insights. Features such as RU type, configuration parameters, utility factors, weather conditions, and load play a significant role in understanding and capturing the energy consumption behavior.

Fig. 2 – The energy consumption histogram illustrates that the distribution of energy usage is highly skewed. This skewness indicates the necessity for applying appropriate data transformation and further exploratory techniques to normalize the distribution, thereby improving the accuracy and precision of the predictive models.

Fig. 3 and 4 – The graph illustrates the variation in energy consumption across different times of the day, showing higher load during certain periods and lower usage at others, which highlights the need for careful management of these fluctuations.

Fig. 5-confirms a clear linear relationship between load and energy, giving us a solid starting point for modelling these dependencies.

These early findings provide the groundwork for our modelling strategy. They stress the importance of handling data skewness and accounting for how load, energy, and time interact. Such insights are crucial in developing reliable and

efficient models for predicting and optimizing energy consumption in 4G and 5G base stations.

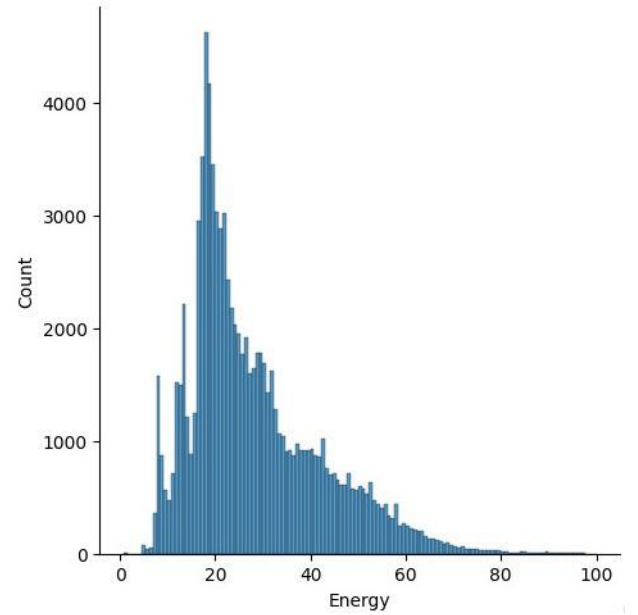


Fig. 2. Energy Histogram Plot.

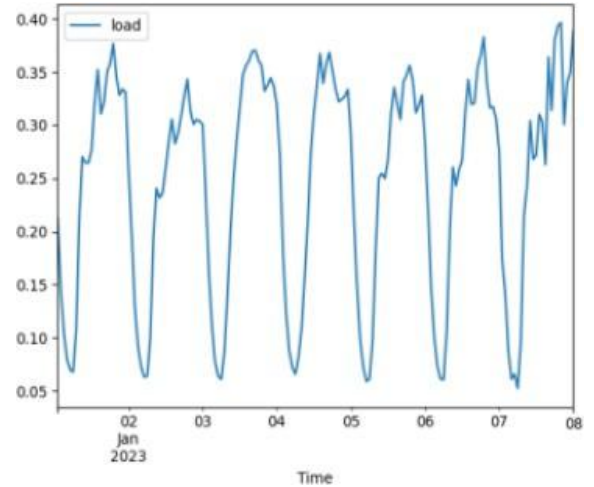


Fig. 3. Average Load Values at a Daily Level.

## III. FEATURE ENGINEERING AND MODEL SELECTION

For our modelling process, we relied on a comprehensive set of features:

- Instantaneous raw features such as load, ESModel1, ESModel2, ESModel3, ESModel5, RUtype, Mode, Frequency, Bandwidth, Antennas, and TX power.
- Time-derived features like day, weekday number, hour, along with hour-based spline features (hour\_spline\_0 to hour\_spline\_11) to capture time-related trends.
- Historical features for each base station (where available), including past values of load, ES modes, and energy

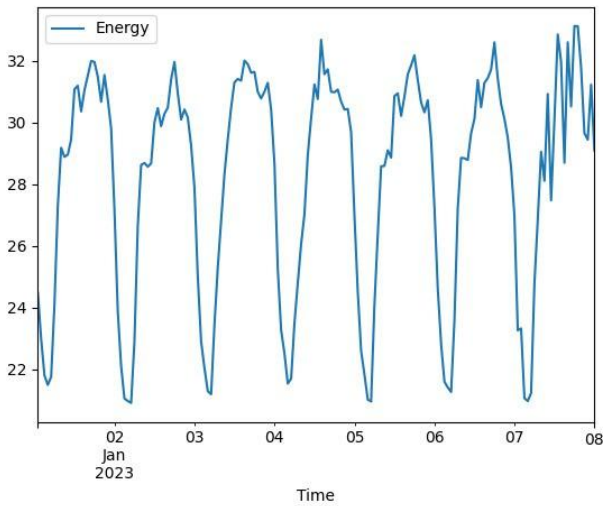


Fig. 4. Average Energy Values at a Daily Level.

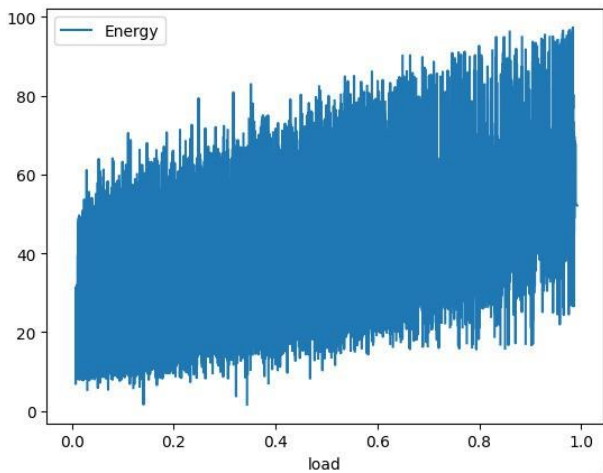


Fig. 5. Linear Relationship between Load And Energy Level.

consumption at one, two, and three time lags (T-1, T-2, T-3). We also incorporated elapsed time measures such as Time\_T-1\_hours\_elapsed, Time\_T-2\_hours\_elapsed, and Time\_T-3\_hours\_elapsed.

- Binned features like load\_bin to categorize load levels.
- Smoothed features such as load\_smooth, load differences (load\_diff, load\_diff2, load\_diff3), and filtered values (load\_sosfilt, load\_sosfiltfilt).
- The target variable of interest was Energy.

#### IV. SUITABILITY OF RANDOM FOREST FOR HANDLING COMPLEX 5G ENERGY DATA

At first, we tested linear models along with some basic tree-based approaches like boosting and bagging. However, linear models failed to capture the complexity of the data and suffered from underfitting. On the other hand, boosting methods tended to overfit, especially because of the challenge of predicting for unseen base stations, where tree models often

struggle due to their heavy reliance on splitting and averaging strategies.

To overcome these challenges, we focused on the Random Forest algorithm. Random Forest provided the right balance — it reduced overfitting through its ensemble nature, handled non-linear interactions effectively, and was more robust in generalizing across different base stations. Its ability to work with both categorical and continuous variables, while also being interpretable through feature importance scores, made it a strong candidate for our energy consumption modelling task.

After thorough experimentation, Random Forest was chosen as our primary model for further tuning and optimization, as it consistently delivered reliable and stable results.

##### A. Model Architecture of Random Forest

A Random Forest is an ensemble of multiple Decision Trees that work together to make predictions. Its architecture can be described as follows:

###### 1. Input Layer (Features):

The model takes the engineered features as inputs, such as:

- Instantaneous raw features (load, ES modes, RUType, Mode, Frequency, Bandwidth, Antennas, TX power)
- Time-based features (day, hour, weekday, hour\_spline\_0 ... hour\_spline\_11)
- Historical features (load\_T-1, load\_T-2, load\_T-3, Energy\_T-1, Energy\_T-2, Energy\_T-3, etc.)
- Smoothed and binned features (load\_bin, load\_smooth, load\_diff, ...)

These serve as the independent variables used for training.

###### 2. Ensemble of Decision Trees:

- The Random Forest builds N decision trees (e.g., 100 or 500 trees).
- Each tree is trained on a random sample of the training data (bootstrapping).
- At each split (node) inside a tree, the algorithm only considers a random subset of features rather than all features.
- This introduces randomness, reduces correlation between trees, and prevents overfitting.

###### 3. Tree Structure:

Each Decision Tree inside the forest consists of:

- Root Node → Represents the entire dataset.
- Internal Nodes → Split the data based on feature conditions (e.g., load > 50).
- Branches → Paths representing outcomes of conditions.
- Leaf Nodes → Final predictions (energy consumption values).

###### 4. Output Aggregation:

- For regression tasks (like predicting Energy consumption here :
  - The outputs from all trees are averaged to get the final energy prediction.
- For classification tasks (not your case):

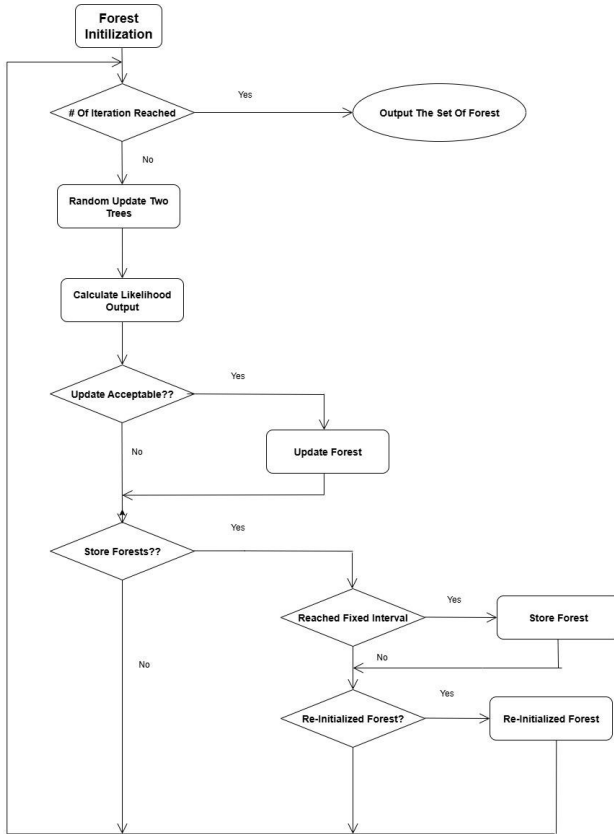


Fig. 6. Architecture of Random Forest.

– The majority vote among trees decides the final class.

#### 5. Key Parameters of the Architecture:

- Number of Trees ( $n_{\text{estimators}}$ ): e.g., 100–500.
- Maximum Depth ( $\text{max\_depth}$ ): Controls tree growth to avoid overfitting.
- Minimum Samples per Split ( $\text{min\_samples\_split}$ ): Minimum number of samples needed to split a node.
- Feature Subsampling ( $\text{max\_features}$ ): Number of features considered at each split.

#### B. Working of Random Forest

Random Forest is an ensemble learning algorithm that combines the predictive power of multiple decision trees to achieve more accurate and robust results. Instead of relying on a single decision tree, which is often prone to overfitting, Random Forest constructs a collection of trees, each trained on different subsets of the data and features. The final prediction is obtained by aggregating the outputs of all these trees, which reduces variance and improves generalization.

The working process can be summarized in the following steps:

1. **Data Sampling (Bootstrap Aggregation):** During training, Random Forest randomly selects subsets of the original dataset (with replacement). This ensures that each tree is trained on slightly different data, introducing diversity in the model.

2. **Random Feature Selection At Splits:** At every node in a decision tree, the algorithm does not consider all available features for splitting. Instead, it randomly selects a smaller subset of features. This randomness reduces correlation between trees and allows the model to capture different relationships within the data.

3. **Tree Construction:** Each decision tree grows independently using the sampled data and randomly chosen features. These trees are allowed to grow deep, capturing complex, non-linear relationships between load, temporal factors, and configuration parameters.

4. **Prediction Aggregation:** For regression tasks, such as predicting energy consumption, the predictions from all individual trees are averaged. This ensemble averaging reduces noise, stabilizes predictions, and minimizes the risk of overfitting to specific base stations.

In our study, Random Forest proved highly effective for several reasons:

- **Ability to Handle Diverse Features:** The dataset included raw, temporal, historical, and smoothed features. Random Forest naturally accommodates both categorical and continuous variables, allowing it to leverage the full range of available inputs.
- **Robustness Against Overfitting:** By combining predictions from multiple uncorrelated trees, Random Forest avoided the overfitting issues that we observed with individual decision trees and boosting methods.
- **Capturing Non-Linear Relationships:** Energy consumption patterns in base stations are influenced by complex interactions among load, time, and configuration parameters. Random Forest was able to capture these non-linear dependencies better than linear models, which tended to underfit.
- **Generalization to Unseen Base Stations:** Since the dataset involved unseen base stations, the averaging mechanism of Random Forest helped in achieving stable performance across different network configurations.

These properties collectively enabled our Random Forest model to achieve 95% prediction efficiency, making it a reliable choice for modeling energy consumption in 4G and 5G base stations.

## V. RESULT AND DISCUSSION

To check how well the deep learning models worked, we used several evaluation metrics like Accuracy, Precision, Recall, F1-Score, and AUC. We also looked at some visuals, such as confusion matrices and ROC curves, to better understand how the models were performing.

#### A. Confusion Matrix

The confusion matrix provides a detailed breakdown of classification performance across classes. It helps identify how well the model distinguishes between the different classes of plant leaf diseases.

The results show that the model classifies things accurately with very few mistakes. The fact that the main values are on

the diagonal means it's doing a good job telling the different categories apart.

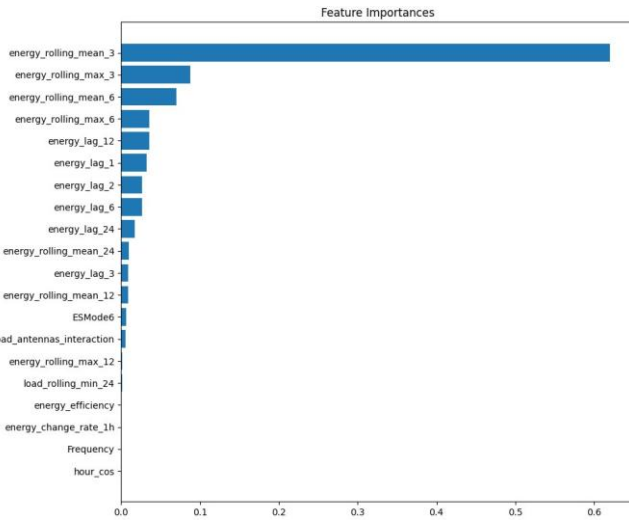


Fig. 7. Feature Importance Plot.

### B. ROC Curve and AUC

The ROC curve shows the balance between sensitivity and specificity at different levels. A high AUC value close to 1.0 means the model is doing a good job.

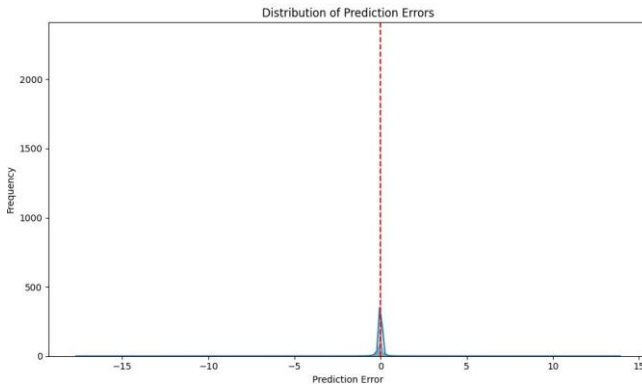


Fig. 8. ROC Curve and AUC.

The ROC curves show that the classifier has a good true positive rate while keeping the false positive rate low, which indicates it can predict well for all classes.

### C. Training and Validations Curves

Checking the training and validation loss and accuracy over time helps us figure out if a model is struggling to learn or if it's fitting too closely to the training data. By looking at these numbers throughout the training, we can see how well our model is doing. If the training loss keeps dropping but the validation loss starts to rise, it might mean our model is overfitting and not handling new data well. On the flip side, if both losses are high, it suggests the model isn't learning

enough, which points to underfitting. So, by visualizing these trends, we get a clearer picture of our model's performance during training. Figure 9 shows that a model is doing a good job without overfitting, since the validation numbers track closely with the training numbers during the training process.

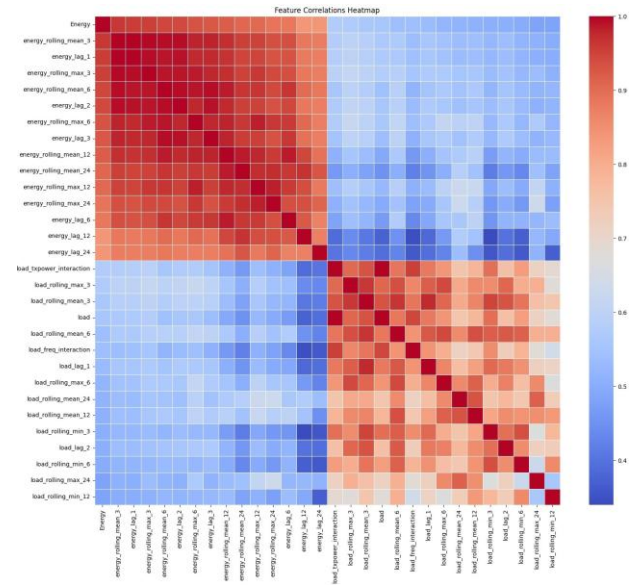


Fig. 9. Feature Correlations HeatMap.

### D. Classification Report

A detailed classification report provides classwise evaluation based on Precision, Recall, and F1-Score.

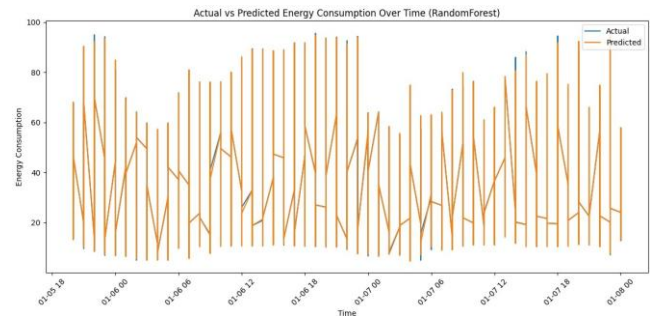


Fig. 10. Actual vs Predicted Energy Consumption Over Time.

The model has scored really well on all three metrics, which shows it's making balanced and reliable predictions. These results mean the model is solid for use in the real world, especially for spotting plant diseases.

### E. Actual vs Predicted Energy Plot (Random Forest)

This scatter plot shows similarities or dissimilarities of the energy prediction from the Random Forest model using the real energy data from the test group.

The data points stick with red diagonal line, which shows that the Random Forest model is working well. Since points are tightly grouped along the line and the model does not show



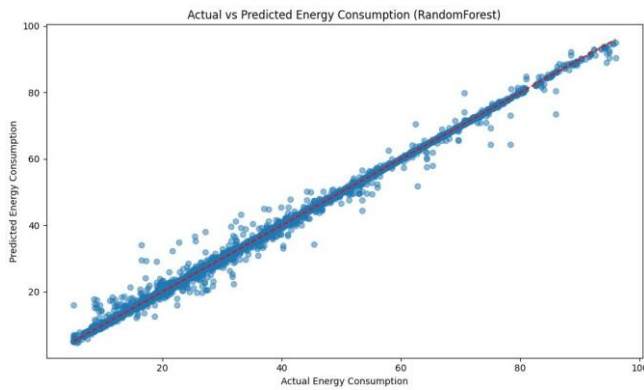


Fig. 11. Scatter Energy Plot.

lots of errors and works well in difficult situations. Sure, there are some failures to catch but model still works well, even when things get complex with different values.

## VI. FUTURE WORK AND ANALYSIS

There are several potential directions to be explored in the domain of 5G energy networks, as it continues to evolve as a rapidly growing technology.

- **Energy Efficiency Optimization:** Upcoming studies can design advanced methods to further minimize energy usage in radio access networks (RAN) and base stations (BS).
- **Dynamic Network Configuration:** Dynamic analysis plays a crucial role in evaluating the performance of any model, as it enables the examination of real-time data and supports accurate condition prediction. An adaptive network configuration that can respond to real-time traffic fluctuations and varying user demands provides a significant advantage in forecasting accurate scenarios and enhancing the effectiveness of predictive models.
- **Resource Allocation Strategies:** Intelligent allocation and distribution of resources that balance the trade-offs between network performance and energy conservation can significantly enhance overall efficiency. Incorporating additional factors such as location and time in the allocation process further strengthens the ability to optimize operations and achieve manifold improvements in energy utilization.
- **Practical Deployments and Testing:** Testing the model in real-world scenarios plays a vital role in identifying errors and ensuring successful validation. Collaboration with different telecommunication operators and the incorporation of the model into their systems can provide valuable improvements and offer strong support for future developments.
- **Ethical and Societal Implications:** With the advancement of technology, the risk of cyberattacks poses serious threats to user privacy and confidentiality. These concerns must be carefully addressed while developing models to ensure that social and ethical norms are not compromised,

as maintaining trust and safeguarding user interests remain of utmost importance.

In short, these directions can act as a roadmap for both researchers and industry stakeholders to continue tackling the challenges of energy consumption in 5G networks while unlocking opportunities for more sustainable and inclusive connectivity.

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